

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA0510

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks 50

Duration :60 Minutes

Annexure A

Scenario-Based Questions

Code of Conduct:

I M V Veeranjaneya Reddy (192472140) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M V Veeranjaneya Reddy

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	

Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

Scenario-Based Questions

1. Low Contrast Image

A grayscale image appears dull with very little difference between object and background intensities.

- Clearly interpret the scenario and explain the cause of low contrast.
- Identify all key issues affecting image visibility.
- Apply an appropriate enhancement technique to address the problem.
- Justify your choice with logical reasoning.
- Present your explanation in a clear and structured manner.

ANS:-

Low Contrast Image

(a) Interpretation of the Scenario and Cause of Low Contrast

A grayscale image looks dull because the pixel intensity values are concentrated within a small range instead of using the full intensity range (0–255). As a result, the object and background have nearly similar brightness, making them difficult to distinguish.

Cause:

- Poor lighting during image capture.
- Limited dynamic range of the camera.
- Incorrect exposure or imaging conditions.

(b) Key Issues Affecting Image Visibility

- Poor separation between object and background.
- Fine details are difficult to observe.
- Edges appear weak and unclear.

- Overall image looks faded and lacks visual quality.

(c) Appropriate Enhancement Technique

Histogram Equalization

Histogram Equalization is used to improve the contrast by redistributing the pixel intensity values over the entire grayscale range.

Steps:

1. Compute the histogram of the image.
2. Calculate the cumulative distribution function (CDF).
3. Map the old intensity values to new intensity values.
4. Generate the enhanced image with improved contrast.

(d) Justification

Histogram Equalization is the most suitable technique because:

- It spreads the intensity values across the full grayscale range.
- It increases the difference between object and background.
- It enhances edges and hidden details.
- It improves overall image visibility without changing the image content.

(e) Structured Explanation

Problem:

Low-contrast grayscale image with poor visibility.

Technique Used:

Histogram Equalization.

Working:

Redistributes pixel intensities using the histogram and CDF to utilize the full intensity range.

Result:

- Better contrast.
- Clearer objects and background.
- Improved edge visibility.
- Enhanced overall image quality.

Conclusion:

Histogram Equalization is an effective contrast enhancement technique for low-contrast grayscale images because it improves visibility by spreading intensity values uniformly, making image details easier to identify.

2. Uneven Illumination

An image captured under non-uniform lighting shows bright and dark patches.

- (a) Interpret the scenario and explain the impact of uneven illumination.
- (b) Identify key factors causing intensity variation.
- (c) Apply a suitable enhancement method to normalize illumination.
- (d) Justify your selected approach with reasoning.
- (e) Provide a clear and well-structured explanation.

ANS:-

Uneven Illumination**(a) Interpretation of the Scenario and Impact**

An image captured under non-uniform lighting contains bright and dark regions. This uneven illumination makes some parts of the image too bright while others are too dark, reducing visibility and making object detection difficult.

Impact:

- Loss of details in bright and dark areas.
- Uneven image appearance.
- Reduced accuracy in image analysis.

(b) Key Factors Causing Intensity Variation

- Uneven lighting or shadows.
- Different light source positions.
- Camera exposure variations.
- Reflections from object surfaces.

(c) Suitable Enhancement Method

Adaptive Histogram Equalization (AHE/CLAHE)

Adaptive Histogram Equalization improves contrast by processing small regions of the image separately. CLAHE (Contrast Limited Adaptive Histogram Equalization) also limits excessive contrast, reducing noise amplification.

Steps:

1. Divide the image into small regions (tiles).
2. Apply histogram equalization to each region.
3. Limit contrast (CLAHE) to avoid over-enhancement.
4. Combine the regions to obtain a uniformly enhanced image.

(d) Justification

CLAHE is the best choice because:

- It enhances local contrast in both bright and dark regions.
- It reduces the effects of uneven illumination.
- It preserves image details.
- It prevents excessive noise amplification.

(e) Structured Explanation

Problem:

Image has bright and dark patches due to uneven lighting.

Technique Used:

Contrast Limited Adaptive Histogram Equalization (CLAHE).

Working:

Enhances contrast locally by equalizing small image regions while limiting over-enhancement.

Result:

- Uniform brightness.
- Better visibility of details.
- Improved local contrast.
- Enhanced overall image quality.

Conclusion:

CLAHE is an effective method for correcting uneven illumination because it enhances local contrast, normalizes brightness across the image, and preserves important details without introducing excessive noise.

3. Salt-and-Pepper Noise

A scanned document contains salt-and-pepper noise affecting readability.

- (a) Explain the scenario and characteristics of this noise.
- (b) Identify key issues impacting image quality.
- (c) Apply an appropriate filtering technique.
- (d) Justify why this method is more suitable than alternatives.
- (e) Present your response clearly and logically.

ANS:-

Salt-and-Pepper Noise

(a) Scenario and Characteristics

A scanned document contains **salt-and-pepper noise**, which appears as random **white (salt)** and **black (pepper)** pixels. This noise is usually caused by scanning errors, dust, or transmission faults and reduces the readability of the document.

Characteristics:

- Random black and white dots.
- Affects only some pixels.
- Makes text and image details unclear.

(b) Key Issues Affecting Image Quality

- Reduced readability of text.
- Loss of important details.
- Unwanted black and white spots.
- Poor visual quality.

(c) Appropriate Filtering Technique

Median Filter

The **Median Filter** is the most suitable method for removing salt-and-pepper noise.

Steps:

1. Select a small neighborhood (e.g., 3×3 window).
2. Arrange the pixel values in ascending order.
3. Replace the center pixel with the median value.
4. Repeat for the entire image.

(d) Justification

The Median Filter is preferred because:

- It effectively removes salt-and-pepper noise.
- It preserves image edges better than averaging filters.
- It removes isolated noisy pixels without blurring the image.
- It improves document readability.

(e) Structured Explanation

Problem:

Scanned document contains salt-and-pepper noise.

Technique Used:

Median Filter

Working:

Replaces each noisy pixel with the median value of its neighboring pixels.

Result:

- Removes black and white noise.
- Preserves edges and text.
- Improves readability.
- Produces a cleaner image.

Conclusion:

The **Median Filter** is the best choice for salt-and-pepper noise because it removes impulse noise while preserving edges and fine details, making the scanned document clear and easy to read.

4. Blurred Edges

An image processed for noise removal appears overly smooth, causing edges to blur.

(a) Interpret the scenario and explain why edges are blurred.

(b) Identify key issues related to smoothing and edge loss.

(c) Apply an appropriate technique to restore edges.

(d) Justify your decision with clear reasoning.

(e) Structure your explanation clearly.

ANS:-

Blurred Edges

(a) Interpretation of the Scenario

After noise removal, the image appears overly smooth because a smoothing filter has reduced both noise and important edge information. As a result, object boundaries become blurred and fine details are lost.

Cause:

- Excessive smoothing using filters such as the mean or Gaussian filter.
- High-frequency components (edges) are reduced along with noise.

(b) Key Issues Related to Smoothing and Edge Loss

- Blurred object boundaries.
- Loss of fine details.
- Reduced image sharpness.
- Difficulty in detecting edges and features.

(c) Appropriate Technique

Laplacian Sharpening Filter

The **Laplacian filter** is used to restore and enhance edges by emphasizing high-frequency components.

Steps:

1. Apply the Laplacian operator to detect edges.
2. Add the Laplacian result to the original image.
3. Obtain a sharper image with clearer edges.

(d) Justification

The Laplacian filter is suitable because:

- It enhances blurred edges.
- It restores fine details lost during smoothing.
- It improves image sharpness.
- It highlights high-frequency components without changing the overall image structure.

(e) Structured Explanation

Problem:

Image becomes overly smooth after noise removal, causing blurred edges.

Technique Used:

Laplacian Sharpening Filter

Working:

Detects edges using the Laplacian operator and combines them with the original image to increase sharpness.

Result:

- Sharper edges.
- Better visibility of details.

- Improved image clarity.
- Enhanced overall quality.
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Conclusion:

The **Laplacian Sharpening Filter** is an effective technique for restoring blurred edges because it enhances high-frequency components, making object boundaries and fine details clear while improving image sharpness.

5. High-Frequency Noise

A satellite image contains fine-grained noise affecting feature visibility.

- Interpret the scenario and describe high-frequency noise.
- Identify the key issues affecting image clarity.
- Apply an appropriate frequency domain filtering method.
- Justify the choice of filter with reasoning.
- Present your answer in a clear and organized manner.

ANS:-

High-Frequency Noise

(a) Interpretation of the Scenario

A satellite image contains **high-frequency noise**, which appears as fine-grained random variations in pixel intensity. This noise hides important features such as roads, buildings, and land boundaries, reducing image quality.

Characteristics:

- Random fine-grained noise.
- Affects high-frequency components.
- Reduces visibility of important features.

(b) Key Issues Affecting Image Clarity

- Reduced image clarity.
- Important features become difficult to identify.
- Lower image quality.
- Decreased accuracy in image analysis.

(c) Appropriate Frequency Domain Filtering Method

Low-Pass Filter (LPF) – Butterworth Low-Pass Filter

A **Butterworth Low-Pass Filter** is used to remove high-frequency noise while preserving most of the useful image information.

Steps:

- Convert the image to the frequency domain using the Fourier Transform.
- Apply the Butterworth Low-Pass Filter to suppress high-frequency components.
- Perform the Inverse Fourier Transform to obtain the filtered image.

(d) Justification

The Butterworth Low-Pass Filter is suitable because:

- It effectively removes high-frequency noise.
- It provides smoother results than an Ideal Low-Pass Filter.
- It preserves important image details better.
- It minimizes ringing effects while improving image quality.

(e) Structured Explanation

Problem:

Satellite image contains fine-grained high-frequency noise.

Technique Used:

Butterworth Low-Pass Filter (Frequency Domain)

Working:

The filter reduces unwanted high-frequency components in the frequency domain and reconstructs the image using the inverse Fourier transform.

Result:

- Reduced noise.
- Clearer image.
- Better visibility of features.
- Improved overall image quality.
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Conclusion:

The **Butterworth Low-Pass Filter** is an effective frequency domain technique for reducing high-frequency noise because it suppresses unwanted noise while preserving important image features and producing a smoother, clearer image.